Alloy-XRD-Bench: A Hierarchical Benchmark for Site Occupancy, Chemical Ordering, and Overlapping-Peak Decomposition

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Abstract

Accurate reconstruction of a powder X-ray diffraction (XRD) pattern does not by itself establish correct recovery of its latent structure. We present Alloy-XRD-Bench, a first-stage benchmark toward alloy powder-XRD refinement. Its two complementary subproblems are HEA-SRO for short-range chemical-order regression in high-entropy alloys, and Gamma-Decomp (gamma/gamma-prime decomposition) for recovering unordered, phase- and reflection-specific components from overlapping peaks. The release contains 84,019 structure–XRD pairs, 10,000 mixtures, and 21,116 overlap windows, with traceable structure–reflection–component–observation provenance. Same-composition controls and grouped in-distribution (ID) and out-of-distribution (OOD) splits test shortcut use and generalization. Task-specific metrics evaluate parent-macro SRO error and matched component recovery separately from whole-pattern reconstruction; no cross-track score is used. Composition-only, peak-feature, and neural baselines provide complementary reference points rather than assuming that a more complex model must perform better. After correcting zero-count SRO targets and retraining, the Spectrum Transformer achieves an in-distribution parent-macro mean absolute error (MAE) of 0.4156 versus 0.5310 for Composition Ridge. A same-learner feature ablation reduces error by 19.90–22.56% when adding peak descriptors to composition across four frozen splits. These clean-simulation gains do not establish direct experimental SRO observability; fixed-cell and perturbation diagnostics expose important limitations. The revised Gamma evaluation contract is documented, but its complete corrected result matrix remains pending. The current release provides computational latent labels; paired experimental SRO and independently annotated component targets are not included. Its experimental patterns support phase-level and whole-pattern checks. These resources prepare data and evaluation protocols for future refinement models, rather than demonstrating a complete refinement pipeline. Code and benchmark protocols are available at https://github.com/guohajining-cmd/alloy-xrd-bench.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Mathematics of computing** → *Probability and statistics.*

Keywords

X-ray diffraction, short-range order, high-entropy alloys, Ni-based superalloys, inverse problems, out-of-distribution generalization

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1 Introduction

Powder XRD is non-destructive, scalable, and sensitive to long-range crystal structure, but its inverse problem is underdetermined: a measured one-dimensional pattern aggregates many reflections and can be compatible with multiple latent structures. This matters when the target is local chemical order or phase-resolved components rather than a coarse phase label.

The motivating failure case is concrete. In the public high-entropy-alloy (HEA) training split, two $Co_{0.25}Cr_{0.25}Fe_{0.25}Ni_{0.25}$ face-centered cubic (FCC) structures have XRD cosine similarity 0.9967 but corrected first/second-shell short-range-order (SRO) L_1 distance 26.67 (Figure 2). We retain the original released pair; we do not reselect it after correcting the labels. A model can fit the dominant Bragg pattern and still miss the structure-derived local-order target. Similarly, a small residual for a γ/γ' mixture does not establish correct assignment to phase and Miller-index (HKL) components. Alloy-XRD-Bench distinguishes observation matching from recovery of generated latent targets, not simulated consistency from experimentally proven truth.

Two alloy settings motivate complementary tests. High-entropy alloys (HEAs) extend alloy design to multiple principal elements [2, 50]. Their structure, phase behavior, and modelling are reviewed in [12, 14, 32, 36, 48]. Experimental SRO studies include microscopy observations in FCC CrCoNi [52] and atom-probe-based quantification [19]. These studies establish the existence of experimental SRO research, not unique SRO recovery from conventional powder XRD. Our same-composition controls test that distinction under a declared forward model. Ni-based superalloy metallurgy provides the second setting [9, 37, 38, 41]: the released mixtures test separation of related γ/γ' reflection profiles under simulated broadening and acquisition perturbations.

The shared long-term objective is alloy powder-XRD refinement. This benchmark is a first step toward that objective, providing data, targets, and evaluation protocols for future refinement-oriented models. HEA-SRO and Gamma-Decomp address complementary subproblems, local chemical-order information and reflection-resolved components. They are scored separately at this stage; we do not present a completed joint refinement algorithm. Alloy-XRD-Bench separates observational fit from latent correctness and contributes (1) a shared structure–reflection–component–observation hierarchy, (2) same-composition HEA controls and

Table 1: Representative resources and released supervision. "Missing" denotes absence from that resource's contract, not a defect.

Resource	Released evidence	Missing for this benchmark
COD [15]	Open crystal structures	Component truth and grouped inverse evaluation
Materials Project [20]	Computed structures/properties	Observed-pattern component provenance
RRUFF [23]	Mineral XRD, indexing	SRO and pre-mixing component truth
HEA source [27]	Structures, attributes, initial SRO	XRD tasks, perturbations, evaluator
Alloy-XRD-Bench	SRO, reflections, components, XRD, OOD	Independent experimental SRO/component labels

traceable γ/γ' mixtures, (3) task-specific permutation-invariant metrics, and (4) grouped ID/OOD protocols, identifiability diagnostics, and parent-aware uncertainty estimates. Strength and creep-life prediction are outside the benchmark contract.

2 Related Work and Positioning

Classical powder-diffraction procedures [21] and Rietveld refinement [39, 51] provide the measurement and whole-pattern modelling background. Pseudo-Voigt line shapes model diffraction broadening [43], while crystallography packages support standardized profile refinement [28, 45]. Small profile residuals alone do not establish a correct structural model [44]. Warren–Cowley coefficients summarize conditional neighbor probabilities without assigning arbitrary atom indices [5].

XRD learning has addressed symmetry classification, augmentation, and multiphase identification or quantification [16, 25, 35, 42, 46], while set prediction supports unordered component outputs [3, 24]. Confidence calibration [18] and proper probabilistic scoring [13] concern the quality of uncertainty estimates. Domain-generalization benchmarks and uncertainty-under-shift evaluations motivate assessing controlled shifts beyond random-split accuracy [17, 22, 34].

Table 1 contrasts the source resources with the supervision needed for this benchmark.

Materials repositories improve access to structures and calculations [1, 4, 8, 40], and frameworks such as AFLOW automate high-throughput calculations [6], but benchmark supervision for latent SRO and pre-mixing phase/HKL components requires additional forward-chain provenance.

Among the resources considered in this study, benchmark-ready experimental data pairing powder-XRD observations with independently quantified, sample-matched SRO targets are rare. This is a scoped assessment, not an exhaustive dataset census. The current release does not include these experimental SRO labels or independently annotated reflection-level components. Its computational targets provide a controlled starting point for future refinement-model development, complementing the experimental studies above.

3 Dataset Overview

Figure 1 summarizes the two data-generation chains.

For latent material state z and acquisition condition n, the release stores

$$z \longrightarrow \mathcal{R}(z) \longrightarrow \{c_k(\theta; z, n)\}_{k=1}^{K} \longrightarrow y(\theta), \tag{1}$$

where $\mathcal{R}$ is the indexed reflection set and c_k is a broadened phase/HKL component. The observation is

$$y(\theta) = \sum_{k=1}^{K} c_k(\theta; z, n) + b(\theta; n) + \epsilon(\theta; n). \tag{2}$$

Stable sample, structure-state, parent-group, reflection, and component IDs link every level to a frozen manifest, configuration, checksum, and provenance row.

Data significance. The release is useful not merely because it contains additional XRD curves, but because it preserves the hidden causes of those curves. HEA-SRO links spectra to site-resolved chemical order, so users can test whether a model recovers local arrangement rather than composition shortcuts. Gamma-Decomp stores phase/HKL components before they are summed into an observed pattern, so users can test whether a method assigns intensity to the correct physical source rather than only reducing residual error. The paired clean, perturbed, OOD, and identifiability views make failure modes measurable: learnability, shortcut use, acquisition sensitivity, component ambiguity, and principled abstention are separated instead of collapsed into one reconstruction score.

Every source row receives a stable ID, canonical structure hash, composition, quality result, and eligibility flags. Derived labels are deterministic, versioned functions of formal structures. Rejected rows remain accounted for. Perturbed views inherit their clean parent's group and cannot cross a split. Thus reconstruction can be scored independently from recovery of z or $\{c_k\}$.

4 HEA-SRO Data

The source corpus contains ordered cells and special quasirandom structures (SQS) with up to 125 atoms from Al, Co, Cr, Cu, Fe, Mn, Ni, and Si [26, 27]. Of 84,024 source rows, 84,019 pass quality checks and five are rejected. Each accepted structure has one immutable clean Cu Kα pattern with 4,001 points. Perturbations are indexed as views, not additional structures.

For species a, b and shell s, the primary target is

$$\alpha_{ab}^{(s)} = 1 - \frac{P(b \mid a, s)}{c_b}, \tag{3}$$

where c_b is the global concentration. For a present central species, a defined neighbor shell, and $c_b > 0$, a zero $a \to b$ count gives $P(b \mid a, s) = 0$ and hence $\alpha_{ab}^{(s)} = 1$; it is not missing data. Absent species and undefined shells remain masked. The corrected overlay adds 669,370 previously omitted targets across 88,019 rows (84,019 source structures and 4,000 counterfactuals), without changing the frozen splits. An independent implementation agrees on 1,819 checked rows. This validates label arithmetic under the declared shell convention, not bulk-alloy SRO or the physical sufficiency of the XRD simulator. Each target remains linked to its finite structure. Exact atom-index recovery is not a task.

Matching composition and topology removes a composition-only distinction between this pair, but does not by itself isolate the scattering mechanism. The source structures are relaxed, so lattice and positional changes may carry correlated information. A separate 1,000-parent, 4,000-pair diagnostic fixes composition, cell, and positions while changing occupancy. Its historical 0/4,000 threshold-exceedance result is conditional on the original uncalibrated perturbation generator (Section C); it is not a universal

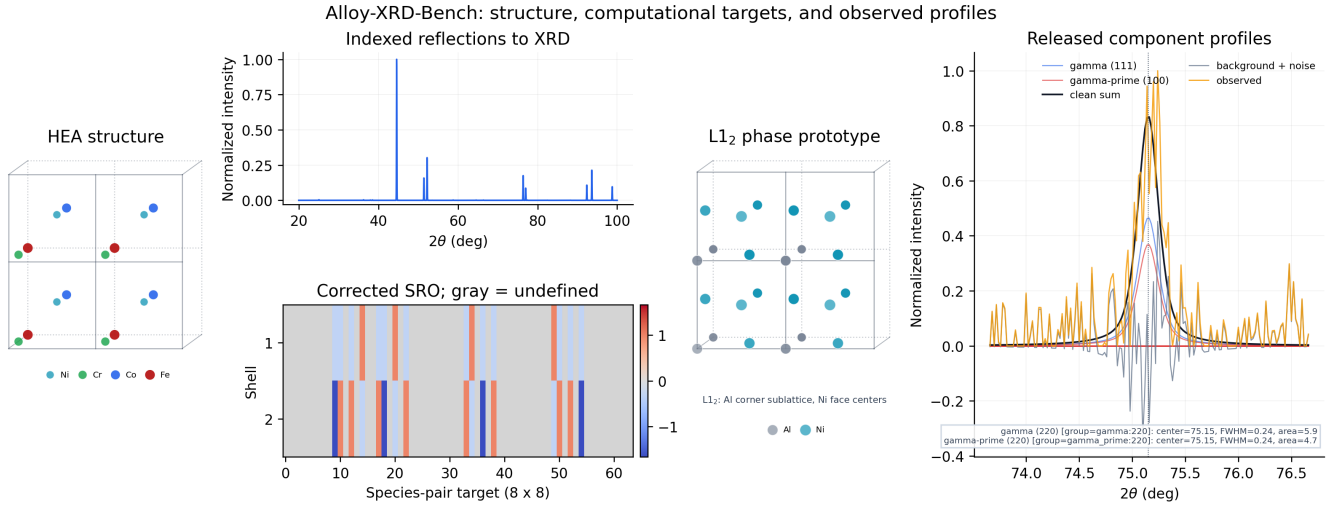


Figure 1: The two released data tracks and their traceable forward chains. Structure panels render projected 2×2 periodic cells for visual readability rather than full simulation supercells; the gamma/gamma-prime panels are phase-prototype guides for FCC γ and ordered $L1_2$ γ'. Displayed spectra, labels, component profiles, IDs, selection metrics, reasons, and input hashes are stored in the visual manifest. The HEA panel uses corrected SRO labels; gray entries are undefined, not zero. Gamma profiles retain the released forward model.

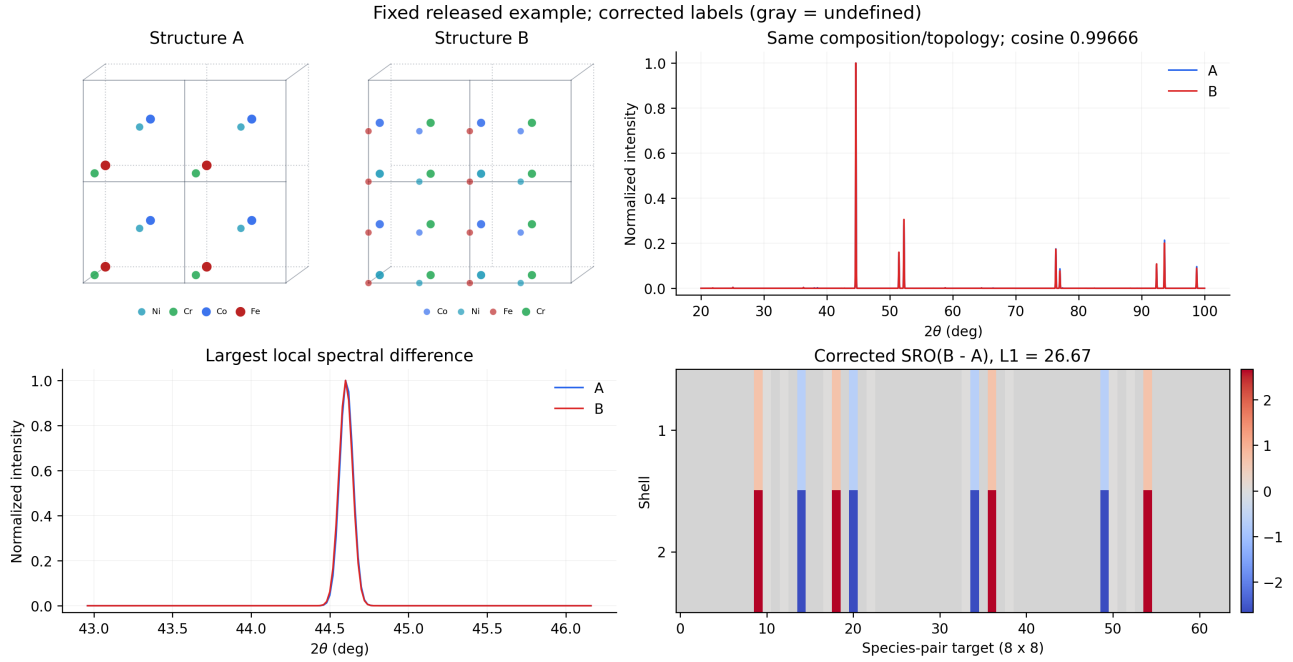


Figure 2: Fixed released HEA training example, now shown with corrected SRO labels. The original selection maximized eligible SRO difference among same-composition/topology pairs with cosine similarity at least 0.995 under the old mask. The pair is not reoptimized here. Corrected L_1 distance is 26.67 across 32 common defined targets; gray entries remain undefined. Structure panels project 2×2 periodic cells.

non-identifiability result. Family labels remain automated diagnostics rather than adjudicated phase truth.

5 Gamma/Gamma-Prime Decomposition Data

Gamma-Decomp is the stable task ID for this γ/γ' track. The generator pairs FCC γ and $L1_2$ γ' structures and varies phase weight, broadening, strain, crystallite size, background, zero shift, and noise within declared ranges. Physically relaxed and parameterized counterfactual pairs are explicitly distinguished. Component truth is stored before summation.

Gamma-Decomp v1.1 contains 10,000 mixtures, 21,116 windows, 175,150 raw reflection-provenance rows, and 165,150 full-pattern scoring components. The window view has 34,983 scoring components. Strictly coincident γ' (221) and (300) reflections remain separate in provenance but merge into an equivalent HKL scoring group. Seventeen phase/HKL groups are evaluated.

Scoring difficulty is computed from eligible targets after merging equivalent components. Context reflections do not define the scoring bucket. For each eligible pair, the separation ratio is

$$\rho_{ij} = |m_i - m_j| / \{(w_i + w_j)/2\},$$

where m is peak center and w is full width at half maximum (FWHM). Zero-target windows are empty; one-target windows are single. For at least two targets, $\rho_{\min} = \min_{i<j} \rho_{ij}$ defines severe if $\rho_{\min} < 0.5$, strong if $0.5 \le \rho_{\min} < 1$, partial if $1 \le \rho_{\min} < 2$, and resolved if $\rho_{\min} \ge 2$. Thus "severe overlap" means peak centers are separated by less than half the average FWHM, not an aesthetic choice made after seeing model errors. Released context buckets are retained as provenance and must not substitute for these target-based buckets.

Figure 3 illustrates how constituent profiles combine in resolved and severely overlapping windows. The target is an unordered component set with phase, equivalent HKL group, center, width, area, and profile parameters. An observed peak may contain several such components. Low reconstruction error therefore does not imply correct latent decomposition; evaluation uses deterministic optimal bipartite matching rather than output order.

6 Coverage, Tasks, and Splits

Figure 4 summarizes release coverage and diagnostic difficulty; Table 2 lists the two task contracts.

The clean reference uses Cu Kα, 20°–100° in 2θ, and 0.02° spacing. Reflection tables retain position, intensity, HKL, and multiplicity. Clean truth is separated from broadening, background, strain, texture, and noise. Validation/test perturbations are frozen.

HEA groups structure hashes, parents, and all observation views before assigning ID, composition OOD, chemistry OOD, and SRO OOD splits. Gamma groups parent pairs and views before instrument and overlap slices. Nearest-neighbor audits cover structure and spectral spaces. An ordinary random split is not presented as evidence of new-chemistry or acquisition generalization.

HEA macro-averages masked MAE within a parent and then across parents. Relative improvement over mandatory Composition Ridge is

$$S_{\mathrm{SRO}} = 100 \max(0, 1 - \mathrm{MAE}_{model}/\mathrm{MAE}_{base}). \quad (4)$$

Per-shell Spearman correlation pools valid targets within each shell; it is not a parent-macro statistic. We retain the original SRO-OOD membership as *legacy SRO OOD*: its label-shift interpretation has not been re-established under the corrected mask. Gamma's protocol requires component F1, joint phase/HKL recovery, count accuracy, and matched center/width/area/profile errors. Its complete revised matrix remains pending. Matching requires equal phase and equivalent HKL, with center error at most $\max(0.15°, 1.5w_{\mathrm{true}})$; the cost combines normalized center, width, and profile errors. Appendix D defines aggregation and empty-target handling. Resampling follows the bootstrap principle [10], with analysis-specific replicate counts stated below. No cross-track composite score is defined because SRO regression and unordered component recovery use different targets.

7 Benchmark Validation

7.1 Corrected HEA Results

All HEA results in Tables 3–4 use corrected labels, frozen original splits, and the original clean observations. Table 3 reports *retraining*, not rescoring of old predictions. Composition Ridge is deterministic; the other formal models record seeds 2027–2031. XGBoost yields identical predictions across these seeds, so its zero standard deviation (SD) does not represent five distinct stochastic fits. Appendix A gives training settings.

Spectrum Transformer has the lowest mean MAE among the four formal models on every split. On ID it improves over Composition Ridge from 0.5310 to 0.4156 (21.73%), while Peak XGBoost reaches 0.4402 and ResNet1D 0.5121. Thus spectral predictors are not interchangeable: a neural architecture alone does not guarantee a large gain, and handcrafted peak descriptors remain competitive. Error is highest on the legacy SRO-OOD membership for every model; this is an empirical difficulty comparison, not renewed validation of that split's SRO-shift semantics. The exploratory physics-constrained variant improves Transformer MAE by only 0.0003–0.0024; its complete metrics are reported separately in Table 5, without a claim of significant superiority.

7.2 Incremental Value Beyond Composition

Comparing different model families cannot isolate the value of the input. We therefore fit the same XGBoost learner with composition alone (C), 98 peak descriptors alone (X), and their concatenation (C+X), using the same splits, training budget, and seed 2027. Table 4 shows the C+X versus C contrast. ID MAE decreases from 0.5230 to 0.4077, with paired difference −0.1153 and 95% parent-bootstrap interval $[-0.1180, -0.1125]$. Reductions span 19.90–22.56% across four splits. The 10,000-resample intervals (bootstrap seed 20261002) condition on one fitted seed; they do not measure training-seed variation. This is a post-review feature ablation, not a preregistered confirmatory test.

These gains establish predictive information beyond composition under the released clean simulator, not a direct diffuse-scattering SRO signature. With composition, cell, and positions fixed, the Transformer reduces error relative to a zero-change predictor by only 4.84–7.44%; C+X does not improve that paired-change target (Table 6). In an additional matched ID diagnostic, the old perturbation generator increases C+X MAE from 0.2751 to 0.6260. The renderer was subsequently corrected to preserve integrated peak area; matched reevaluation is not complete. The old negative result

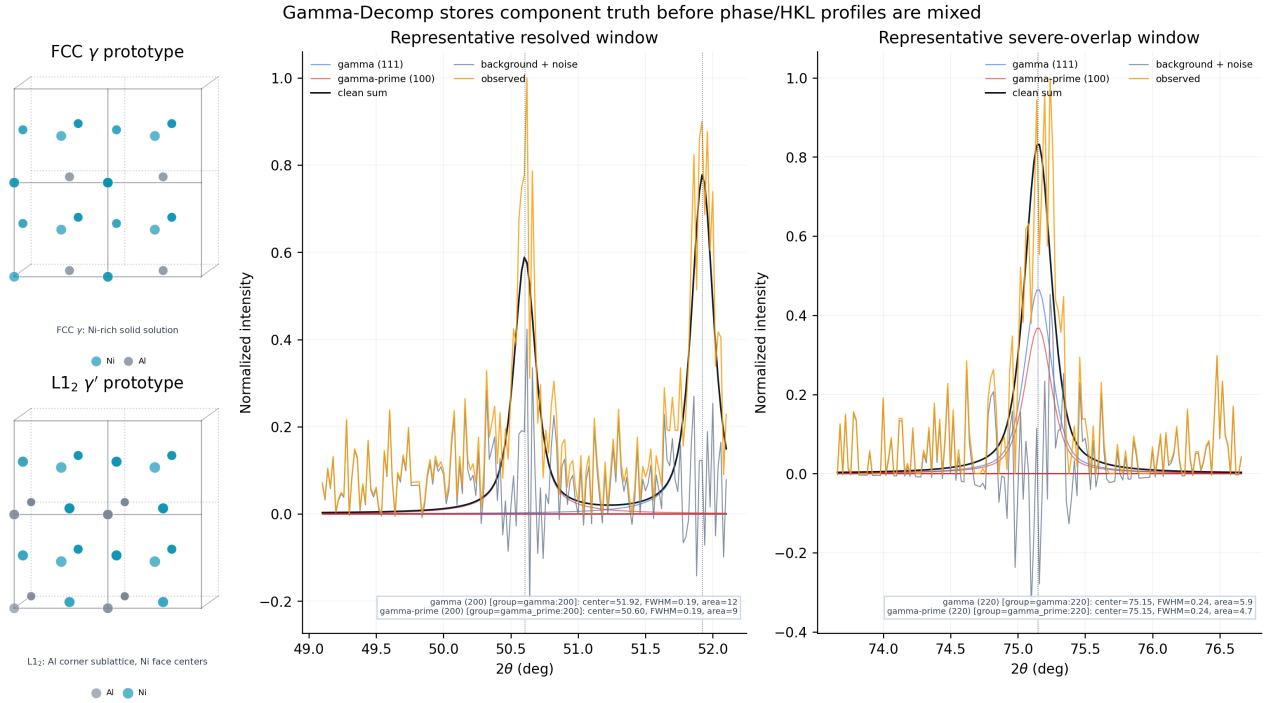


Figure 3: Gamma-Decomp training examples nearest to the resolved and severe-overlap context medians in component count, signal-to-noise ratio (SNR), and phase weight. These are released illustrations, not new corrected-evaluator results. The panels show FCC γ and ordered $L1_2$ γ' prototype guides, phase/HKL profiles, clean sum, background/noise residual, observed pattern, center, FWHM, and area.

Table 2: Dataset and task contract. The two subproblems serve a shared future-refinement objective and are scored separately. Reconstruction is supplementary and never replaces a latent metric.

Track	Released scale	Target	Frozen splits	Primary metrics
HEA-SRO	84,019 structure–XRD pairs	Masked first/second-shell Warren–Cowley matrices	ID; composition, chemistry, SRO OOD	Parent-macro masked MAE; per-shell Spearman
Gamma-Decomp	10,000 mixtures; 21,116 windows	Unordered phase/HKL components and peak parameters	Instrument ID/OOD; resolved, partial, severe	Matched component F1; phase/HKL and count accuracy

Table 3: Corrected-label HEA retraining: parent-macro masked MAE (lower is better), mean ± seed SD. The best formal-model mean in each column is bold, the runner-up underlined, with consistently ranked model names highlighted likewise; this denotes numerical rank, not significance. Ridge has one deterministic fit (no SD shown); the others record five seeds. Legacy SRO OOD retains the original membership but not a revalidated label-shift claim.

Model	ID	Composition OOD	Chemistry OOD	Legacy SRO OOD
Composition Ridge	0.5310	0.5312	0.5167	0.5853
Peak XGBoost	0.4402 ± 0.0000	0.4383 ± 0.0000	0.4332 ± 0.0000	0.5026 ± 0.0000
ResNet1D	0.5121 ± 0.0126	0.5226 ± 0.0079	0.5008 ± 0.0283	0.5557 ± 0.0130
Spectrum Transformer	**0.4156** ± 0.0023	**0.4135** ± 0.0031	**0.4079** ± 0.0013	**0.4859** ± 0.0030

therefore demonstrates sensitivity to that generator, not a calibrated experimental noise limit.

7.3 Gamma Evaluation Status

The revised evaluator distinguishes eligible targets from context, handles empty windows, and removes arbitrary component IDs from physical matching and tied-risk calculations. However, the full prediction inventory, adaptation, and corrected uncertainty analysis are incomplete. Historical Gamma F1 scores are therefore not carried into the updated main table. A constrained peak-profile fit is not a Rietveld refinement baseline; neither a validated Rietveld comparison nor the requested known-count, known-phase,

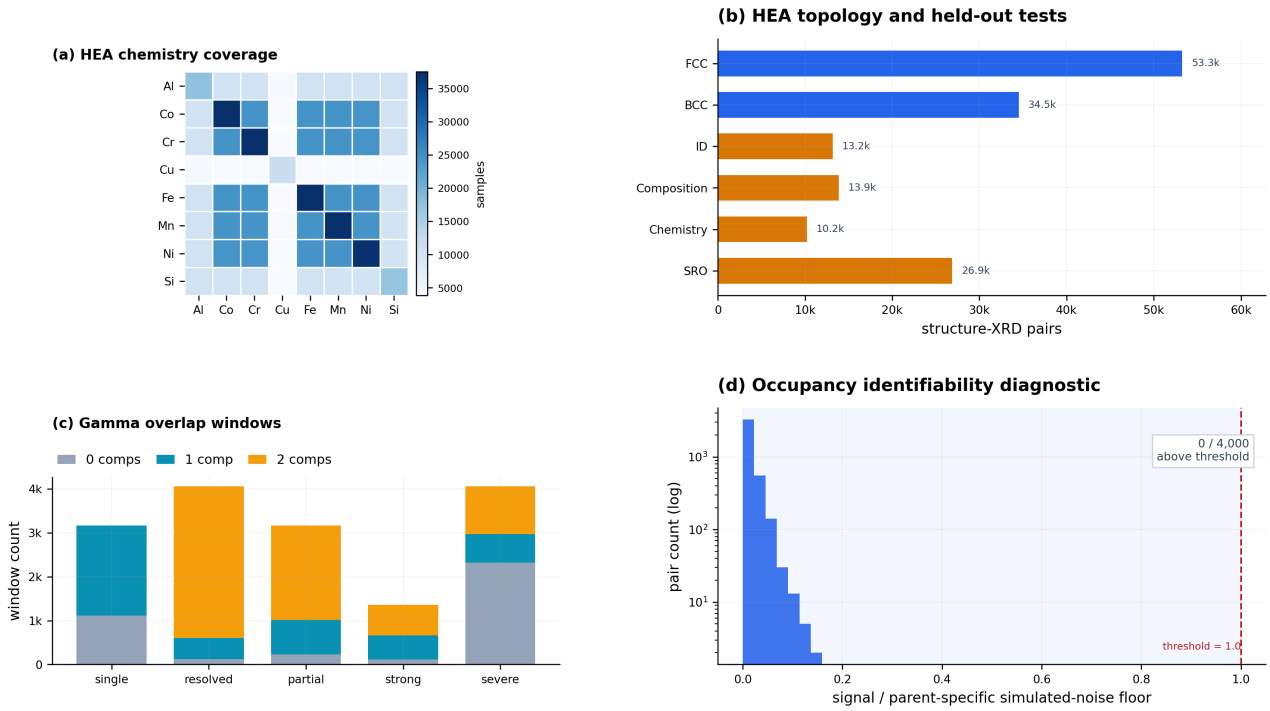


Figure 4: Manifest-derived coverage and difficulty: HEA element co-occurrence, body-centered cubic (BCC)/FCC and ID/OOD support, released Gamma context-count/overlap support, and the historical occupancy identifiability ratio under the old perturbation generator (threshold 1.0). The latter is not experimentally calibrated.

Table 4: Same-learner feature ablation on corrected clean targets: composition (C), peak descriptors (X), and both (C+X). $\Delta = \mathrm{MAE}_{C+X} - \mathrm{MAE}_{C}$; negative favors adding spectral descriptors. Intervals are paired, pointwise, parent-level percentile intervals conditional on the fitted models.

Split	C	X	C+X	Δ MAE	Paired 95% interval	Reduction
ID	0.5230	0.4402	0.4077	-0.1153	[-0.1180, -0.1125]	22.04%
Composition OOD	0.5202	0.4383	0.4029	-0.1173	[-0.1200, -0.1147]	22.56%
Chemistry OOD	0.5049	0.4332	0.3976	-0.1073	[-0.1105, -0.1042]	21.25%
Legacy SRO OOD	0.6005	0.5026	0.4810	-0.1195	[-0.1213, -0.1177]	19.90%

and known-instrument models is available. This limits the current benchmark evaluation, rather than being resolved by the HEA gains.

8 Availability, Limitations, and Conclusion

Code, configuration, evaluation, cards, download tools, and smoke data are released separately from the sharded structures/XRD/labels and the frozen metadata/checksum snapshot. Code is available at https://github.com/guohajining-cmd/alloy-xrd-bench; dataset shards are hosted at https://huggingface.co/datasets/jimi7771/alloy-xrd-bench; and the original v1.1.1 archive is https://doi.org/10.5281/zenodo.21620650. The release also contains the Dataset Card, Benchmark Card, Croissant metadata, citation record, and release checklist, following broader dataset and model reporting practices [11, 33, 49]. The v1.1.2 correction release provides the source-bound SRO-label overlay, revised evaluation code, corrected-result summaries, and this manuscript. Version-specific assets and checksums are available from the GitHub v1.1.2 release and the Hugging Face v1.1.2 dataset revision. The original data and frozen split membership are retained; the correction does not establish completion of the outstanding validation described below.

The computational structure is ground truth for generated samples, but does not prove that experimental powder XRD uniquely determines occupancy. Periodic supercells, similar scattering factors, finite resolution, texture, microstrain, nanostructure, and unmodeled phases can make inversion ambiguous. Family labels are automated and unadjudicated. Occupancy uses simulated perturbations only; real repeated-measurement import and noise calibration are v1.2 gates.

The experimental data included in this release do not provide independent SRO or reflection-level component labels. The HEA experimental set supports mother-phase trends [47]. The Ni-based sanity set uses public raw arrays for heat-treated IN718 and AM718R, where γ' and gamma-double-prime superlattice reflections are not distinguishable [29, 30]. It supports phase-level and whole-pattern checks, not experimental component-level validation. The distinction between γ' and gamma-double-prime is important: Alloy 718 precipitation and dual-superlattice alloy studies concern distinct precipitate structures [7, 31], not interchangeable ground-truth labels for this two-phase benchmark.

The current forward model has not been validated for recovering an explicit diffuse-scattering SRO signal. Finite-cell label arithmetic also does not quantify convergence to bulk order: neither a supercell-size study nor independent configuration-sampling uncertainty is complete. Parent-bootstrap intervals describe predictive-error variation, not these physical or label uncertainties.

Alloy-XRD-Bench provides a first-stage evaluation resource for future alloy powder-XRD refinement models, with complementary local-order and component-decomposition subproblems. Corrected HEA experiments show incremental predictive value over composition on clean simulated observations, but substantially weaker fixed-cell evidence and sensitivity to the old perturbation generator. The contribution is a traceable evaluation resource with explicit failure boundaries, not a new universal inverse-problem principle or demonstrated experimental SRO recovery. Completing corrected Gamma evaluation and external physical validation is necessary before making broader claims about benchmark maturity or experimental transfer. A joint refinement pipeline remains future work.

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A Training and Statistical Protocol

Each model is fit on the training partition of the declared split; neural checkpoint selection uses validation parent-macro MAE. Test observations are not used for checkpoint selection. Networks use masked target loss, AdamW, learning rate 10^{-3}, at most 40 epochs, and patience 6. ResNet1D and Transformer use batch size 128; the exploratory physics variant uses 64, 16 reconstructed peaks, and reconstruction weight 0.1. Ridge uses regularization $\alpha = 1$. Peak XGBoost uses at most 32 peaks, 300 estimators, depth 6, and learning rate 0.05. The feature ablation uses the same tree budget, CPU histogram training, and one seed (2027). Its recorded environment is Python 3.11.15, NumPy 2.4.6, and XGBoost 3.2.0. These settings describe completed experiments, not newly tuned models.

For parent p, we first average absolute error over defined targets of each sample, then over samples of that parent, and finally across held-out parents. The paired bootstrap draws the same parent IDs with replacement for both models, recomputes the mean difference, and takes percentile endpoints. All C versus C+X comparisons use 10,000 replicates and bootstrap seed 20261002. The evidence also stores 98.75% intervals allocating error over these four splits; neither interval family corrects all exploratory contrasts. Across-seed SD and parent-bootstrap intervals answer different questions and are not substituted for each other.

B Complete Corrected HEA Metrics

Table 5 gives every model/split cell for MAE, root mean squared error (RMSE), both shell correlations, and the score relative to the corresponding corrected Composition Ridge baseline. Correlations pool valid entries within each shell; they are not correlations of parent means. The physics variant adds a reconstruction prior and is exploratory, so it is not ranked with the four formal methods in Table 3.

Table 5: Complete corrected-label HEA results, means over recorded seeds. ρ_1, ρ_2 are per-shell Spearman correlations; S_{SRO} is the clipped percentage improvement over Composition Ridge. Bold denotes the lowest MAE on chemistry and legacy SRO OOD, including the exploratory model; no statistical significance is implied.

Split	Model	MAE	RMSE	ρ_1	ρ_2	S_{SRO}
ID	Composition Ridge	0.5310	0.8156	0.227	0.074	0.00
ID	Peak XGBoost	0.4402	0.7017	0.446	0.457	17.09
ID	ResNet1D	0.5121	0.7863	0.194	0.260	3.56
ID	Spectrum Transformer	0.4156	0.6749	0.501	0.514	21.73
ID	Physics Transformer (exploratory)	0.4153	0.6740	0.503	0.514	21.79
Composition OOD	Composition Ridge	0.5312	0.7927	0.243	0.066	0.00
Composition OOD	Peak XGBoost	0.4383	0.6779	0.479	0.472	17.49
Composition OOD	ResNet1D	0.5226	0.7761	0.190	0.245	1.62
Composition OOD	Spectrum Transformer	0.4135	0.6475	0.541	0.538	22.16
Composition OOD	Physics Transformer (exploratory)	0.4111	0.6459	0.545	0.543	22.61
Chemistry OOD	Composition Ridge	0.5167	0.7641	0.240	0.037	0.00
Chemistry OOD	Peak XGBoost	0.4332	0.6593	0.459	0.455	16.16
Chemistry OOD	ResNet1D	0.5008	0.7418	0.254	0.199	3.09
Chemistry OOD	Spectrum Transformer	0.4079	0.6325	0.530	0.526	21.05
Chemistry OOD	Physics Transformer (exploratory)	**0.4069**	0.6316	0.530	0.527	21.26
Legacy SRO OOD	Composition Ridge	0.5853	0.9772	0.089	-0.045	0.00
Legacy SRO OOD	Peak XGBoost	0.5026	0.8605	0.328	0.394	14.13
Legacy SRO OOD	ResNet1D	0.5557	0.9522	0.107	0.175	5.06
Legacy SRO OOD	Spectrum Transformer	0.4859	0.8557	0.367	0.409	16.99
Legacy SRO OOD	Physics Transformer (exploratory)	**0.4847**	0.8508	0.367	0.414	17.19

C Conditional Diagnostics and Their Limits

The fixed-cell evaluation predicts the SRO difference between occupancy variants relative to a zero-change predictor. Table 6 uses only held-out parents: 150 ID, 147 composition OOD, 99 chemistry OOD, and 227 legacy SRO OOD. Transformer averages five fitted seeds; C+X has one. The displayed 4,000-resample intervals condition on those fitted models. These experiments do not supply independently measured SRO.

The released 0/4,000 diagnostic compares occupancy-induced spectral distance with a simulated nuisance floor; the largest ratio is 0.1527 against threshold 1.0. Two pairs have no corrected SRO change. The original angle-domain renderer changes relative integrated peak areas when width changes. Its negative result cannot therefore establish a physical broadening or experimental noise limit. The area-normalized replacement is implemented, but the matched reevaluation required to support revised perturbation claims remains incomplete.

The separate matched ID perturbation comparison uses 150 test parents, the same examples for clean and perturbed views, and one C+X fit. Its MAE increase of 0.3509 has paired 95% interval [0.3328, 0.3693]. Those values are not comparable to the full source-corpus MAEs in Table 4: they describe a different, smaller diagnostic population under the old renderer. Keeping this negative result explicit prevents clean-corpus gains from being mistaken for robust experimental performance.

D Gamma Scoring Contract and Outstanding Evidence

Eligible truth components are grouped by sample. Phase/HKL-gated optimal bipartite matching gives true positives (TP), false positives (FP), and false negatives (FN). Counts are pooled within each parent pair before forming F1, $2\mathrm{TP}/(2\mathrm{TP}+\mathrm{FP}+\mathrm{FN})$; parent scores are then macro-averaged. A zero denominator gives zero F1. The field named phase/HKL accuracy is joint recovery, $\mathrm{TP}/\max(\mathrm{TP}+\mathrm{FN}, \mathrm{TP}+\mathrm{FP}, 1)$, not separate phase and HKL classification accuracy. Count accuracy is the fraction of samples with exactly correct component count within a parent, macro-averaged over parents. Empty-target cases are therefore retained even though their F1 convention differs from count accuracy.

Center, width, integrated-area, and profile errors apply to matched components and must be read alongside F1 to avoid rewarding selective detection. The revised forward-reconstruction reference contains only eligible component curves; it is not the observed whole pattern including context/background. Tied confidence risk uses expected uniform tie permutations. Sorting by component ID is not a physical tie-break.

These definitions are now explicit, but definitions and regression tests do not replace a completed experiment. The outstanding matrix must cover both full patterns and windows, all required metrics, eligible-target difficulty, instrument slices, empty cases, and uncertainty that handles shared structures across parent pairs. Traditional whole-pattern and privileged-information baselines remain missing. No corrected numerical Gamma ranking or superiority claim is asserted in this revision.

Table 6: Fixed-cell paired-change diagnostic relative to zero change. Positive reduction favors the model; negative Δ MAE favors it. Parent-only intervals condition on fitted seeds and are pointwise.

Split	Predictor	Parents	Reduction	95% interval, Δ MAE
ID	C+X	150	-4.27%	[0.0092, 0.0160]
ID	Transformer	150	4.84%	[-0.0166, -0.0121]
Composition OOD	C+X	147	-5.53%	[0.0133, 0.0203]
Composition OOD	Transformer	147	5.11%	[-0.0176, -0.0134]
Chemistry OOD	C+X	99	-6.05%	[0.0141, 0.0227]
Chemistry OOD	Transformer	99	4.95%	[-0.0178, -0.0123]
Legacy SRO OOD	C+X	227	-7.53%	[0.0161, 0.0224]
Legacy SRO OOD	Transformer	227	7.44%	[-0.0216, -0.0163]